The Shark Protocol: Technical Specification & System Architecture

Verifiable, Proof-Carrying AI Execution OS & Security Gateway

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ABSTRACT

As autonomous AI agents acquire tool-calling capabilities and direct access to external APIs, ensuring deterministic runtime alignment becomes a critical imperative. This paper presents The Shark Protocol, a proof-carrying epistemic architecture and security proxy for multi-provider LLM workflows. The system integrates (1) a SHA-256 cryptographic claim verification shell (VRK v0.3.1); (2) a 4D continuous-time ODE cognitive dynamics governor with 17 machine-checked Lean 4 theorems; (3) Low-Rank Spectral Context Memory (LRSC) achieving 88.9% context compression; and (4) a Multi-Cloud Security Gateway supporting Groq, Together AI, Google Gemini, and Hugging Face APIs.

1. System Architecture & Layered Specifications

1.1 Layer 1: VRK Cryptographic Claim Verification Shell

Every agent tool call is wrapped in an immutable schema C = <claim_id, proposition, evidence_class, DAG_deps, authority>. All state transactions are appended to an immutable SHA-256 ledger satisfying h_k = H(h_{k-1} || manifest_k). Uncertified actions trigger immediate security protection.

1.2 Layer 2: 4D ODE Cognitive Dynamics Governor

The state vector x(t) = (gamma, H_obs, I_dot, Psi)^T evolves in 4D phase space under non-linear ODEs. In Lean 4 (Mathlib), 17 machine-checked zero-sorry theorems prove that the trapping box B_4 = [0, gamma_max] x [0, H_max] x [0, I_dot_max] x [0, 1] is invariant under Nagumo subtangentiality.

1.3 Layer 3: Low-Rank Spectral Context Memory (LRSC)

Grounded in the near-Nyquist odd-ring channel operator rank(V) = p, context vectors are projected onto an 11-channel Discrete Fourier Transform (DFT) basis out of 99 candidate frequency bands (r_LRSC = 11/99). Evaluated across 13,432,735,555 subsets, K_0.001 = 11 is the minimum cardinality for relative l_2 error <= 10^-3.

1.4 Layer 4: Multi-Cloud Security Gateway (4 API Integrations)

The gateway runs locally at http://localhost:8000 and proxies traffic across four LLM providers with automatic failover and cryptographic sealing:

Provider	Target Model	Environment Variable	Role / Specialization
Groq API	llama-3.3-70b-versatile	GROQ_API_KEY	Ultra-low latency inference
Together AI	Llama-3.3-70B-Instruct-Turbo	TOGETHER_API_KEY	High-throughput reasoning
Google Gemini	gemini-1.5-flash	GEMINI_API_KEY	Multimodal context synthesis
Hugging Face	Mistral-7B-Instruct-v0.2	HUGGINGFACE_API_KEY	Open-weights fallback

2. Verification Benchmarks & Audit Results

Subsystem	Benchmark Target	Audit Result
Layer 1 (VRK)	Validation Protection Test	33/33 Tests Verified (100%)
Layer 2 (ODE)	Lean 4 Trapping Box Invariance	17 Theorems Checked (0 sorrys)
Layer 3 (LRSC)	Spectral Compression Error	Relative Error <= 10^-3 (1/9th size)
Layer 4 (Gateway)	Multi-API Live Dispatch	4/4 Providers Verified (200 OK)

3. Open-Source Availability

The Shark Protocol is released as open-source software under the MIT License. Complete code, Lean 4 formal proofs, Streamlit dashboard, and pytest suites are available at GitHub: reggaesharkk/The-Reggae-Shark-Protocol.